Week 3: Maximum Likelihood Estimation

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Introduction to Data Science (MSc)

1 Introduction to Maximum Likelihood Estimation

Maximum Likelihood Estimation represents one of the most fundamental and theoretically elegant approaches to parameter estimation in statistical inference and data science. The method provides a unified framework for parameter estimation that combines intuitive appeal with rigorous mathematical foundations, making it indispensable for modern data analysis and machine learning applications.

The theoretical importance of maximum likelihood estimation extends far beyond its computational convenience. The method embodies fundamental principles of statistical inference, including the use of all available information in the data, the principle of parsimony in model selection, and the achievement of optimal statistical properties under mild regularity conditions. Understanding MLE deeply provides essential insights into the foundations of statistical learning and the theoretical underpinnings of many machine learning algorithms.

From a practical perspective, maximum likelihood estimation serves as the foundation for numerous data science methods, from logistic regression and neural networks to hidden Markov models and Gaussian mixture models. The optimization techniques developed for MLE, including the EM algorithm and various numerical methods, have broad applications throughout machine learning and computational statistics.

2 Mathematical Foundations of Likelihood Theory

2.1 Likelihood Function and Its Interpretation

The likelihood function represents a fundamental shift in perspective from probability to statistical inference, transforming the question from "what data might we observe?" to "what parameters best explain the observed data?" This conceptual shift underlies all of statistical inference and provides the foundation for principled parameter estimation.

Definition 2.1 (Likelihood Function)**.** *Let $X_1, X_2, \ldots, X_n$ be a random sample from a population with probability density function $f(x;\theta)$, where $\theta \in \Theta$ is an unknown parameter vector. The likelihood function is defined as:*

$$L(\theta) = L(\theta; x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} f(x_i;\theta)$$

for continuous distributions, or the corresponding probability mass function for discrete distributions.

The likelihood function $L(\theta)$ represents the probability (or probability density) of observing the specific data values $x_1, \ldots, x_n$ as a function of the parameter θ. Crucially, the data are treated as fixed and known, while θ is treated as the variable of interest. This perspective reversal from probability theory enables the quantification of evidence about parameter values based on observed data.

The interpretation of likelihood requires careful attention to its mathematical properties. Unlike probability distributions, likelihood functions are not required to integrate to one over the parameter space. The absolute value of the likelihood is often less important than the relative likelihood of different parameter values, leading to the focus on likelihood ratios in statistical inference.

Definition 2.2 (Log-Likelihood Function)**.** *The log-likelihood function is defined as:*

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta)$$

The logarithmic transformation serves multiple purposes: it converts products to sums (simplifying differentiation), transforms exponential family distributions into linear forms (facilitating optimization), and often improves numerical stability by avoiding underflow problems that can occur with products of small probabilities.

2.2 The Maximum Likelihood Principle

The maximum likelihood principle provides a natural and intuitive approach to parameter estimation: choose the parameter value that makes the observed data most probable. This principle can be formalized mathematically and leads to estimators with desirable theoretical properties.

Definition 2.3 (Maximum Likelihood Estimator)**.** *The maximum likelihood estimator (MLE) of θ is the value $\hat{\theta}_{MLE}$ that maximizes the likelihood function:*

$$\hat{\theta}_{MLE} = \arg\max_{\theta \in \Theta} L(\theta) = \arg\max_{\theta \in \Theta} \ell(\theta)$$

The equivalence between maximizing $L(\theta)$ and $\ell(\theta)$ follows from the monotonicity of the logarithm function. In practice, working with the log-likelihood is almost always preferable due to computational and analytical advantages.

The maximum likelihood principle can be justified from multiple perspectives. From an information-theoretic viewpoint, the MLE minimizes the Kullback-Leibler divergence between the true distribution and the estimated distribution. From a decision-theoretic perspective, the MLE can be derived as the Bayes estimator under certain prior distributions. From an asymptotic perspective, the MLE achieves optimal efficiency properties in large samples.

3 Analytical Solutions and Classical Examples

3.1 Single Parameter Estimation

For many common distributions, the maximum likelihood estimator can be derived analytically by solving the likelihood equation, which is obtained by setting the derivative of the log-likelihood equal to zero.

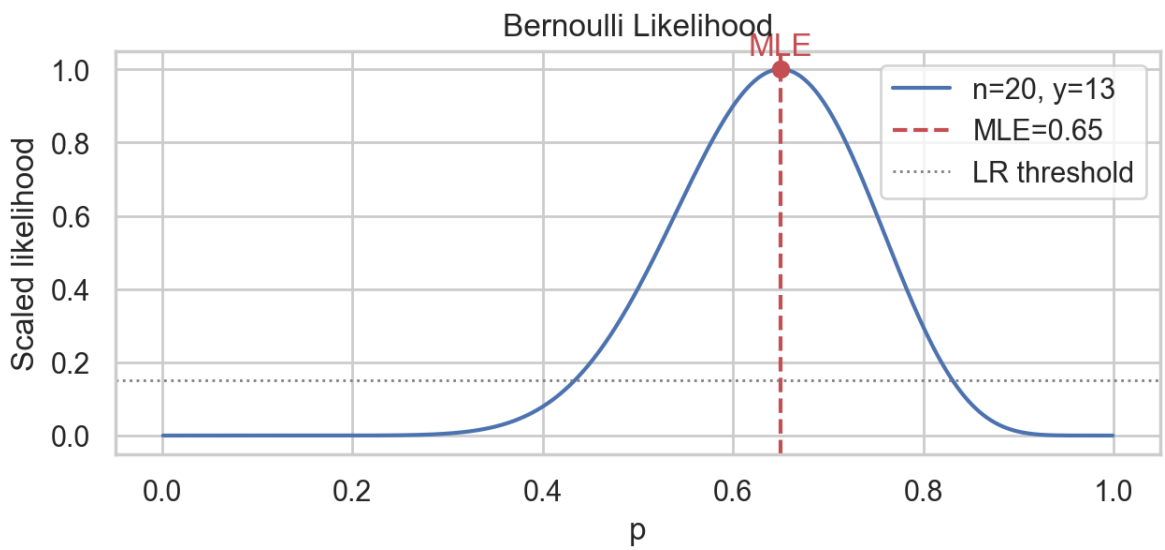

Figure 1: Bernoulli likelihood as a function of p for a dataset with n trials and y successes. The maximum occurs at $\hat{p} = y/n$.

Theorem 3.1 (Likelihood Equation)**.** *If the log-likelihood function $\ell(\theta)$ is differentiable and the maximum occurs in the interior of the parameter space, then the MLE satisfies:*

$$\left.\frac{\partial \ell(\theta)}{\partial \theta}\right|_{\theta=\hat{\theta}_{MLE}} = 0$$

This equation is called the likelihood equation or score equation.

The score function $S(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}$ plays a central role in likelihood theory, representing the rate of change of the log-likelihood with respect to the parameter. The MLE corresponds to the point where the score function equals zero, indicating that the log-likelihood is neither increasing nor decreasing.

Example 3.1 (Normal Distribution with Known Variance)**.** *Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ where σ^2 is known. The log-likelihood function is:*

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

The score function is:

$$S(\mu) = \frac{\partial \ell(\mu)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

Setting $S(\mu) = 0$ and solving yields:

$$\hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$$

The second derivative test confirms this is a maximum:

$$\frac{\partial^2 \ell(\mu)}{\partial \mu^2} = -\frac{n}{\sigma^2} < 0$$

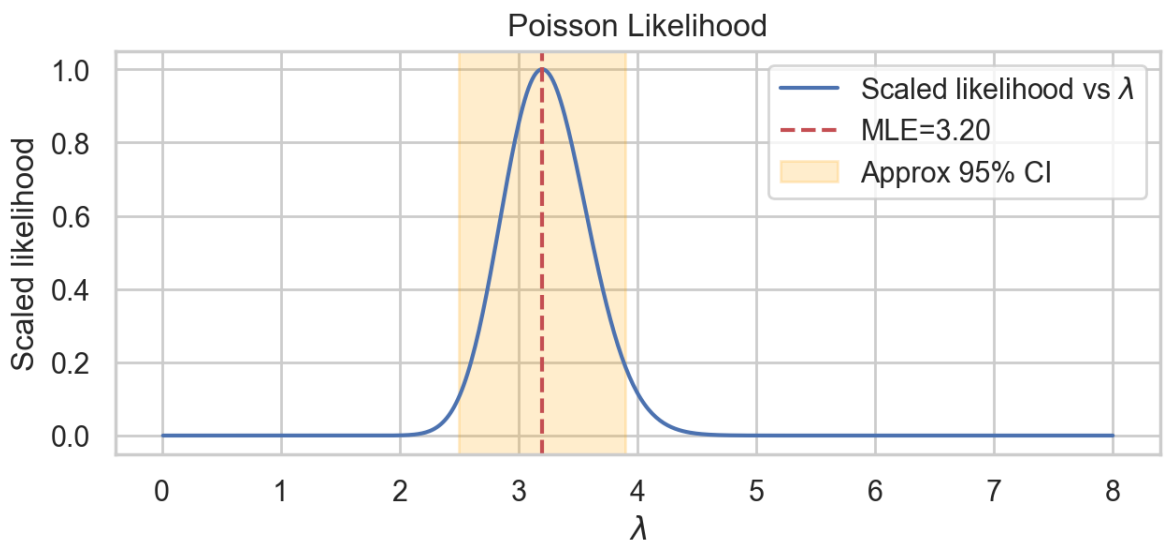

Figure 2: Poisson likelihood as a function of the rate parameter λ. The maximum occurs at $\hat{\lambda} = \bar{Y}$. Shaded band indicates an approximate 95% CI.

This fundamental result shows that the sample mean is the maximum likelihood estimator of the population mean for normal distributions, providing theoretical justification for one of the most basic statistical procedures.

3.2 Multiparameter Estimation

When the parameter vector $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_k)^T$ contains multiple components, the likelihood equations become a system of equations that must be solved simultaneously.

Theorem 3.2 (Multiparameter Likelihood Equations)**.** *For a parameter vector $\boldsymbol{\theta}$, the MLE satisfies the system of equations:*

$$\left.\frac{\partial \ell(\boldsymbol{\theta})}{\partial \theta_j}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_{MLE}} = 0, \quad j = 1, 2, \ldots, k$$

Example 3.2 (Normal Distribution with Unknown Mean and Variance)**.** *Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ where both parameters are unknown. The log-likelihood function is:*

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

The likelihood equations are:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0 \tag{1}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i - \mu)^2 = 0 \tag{2}$$

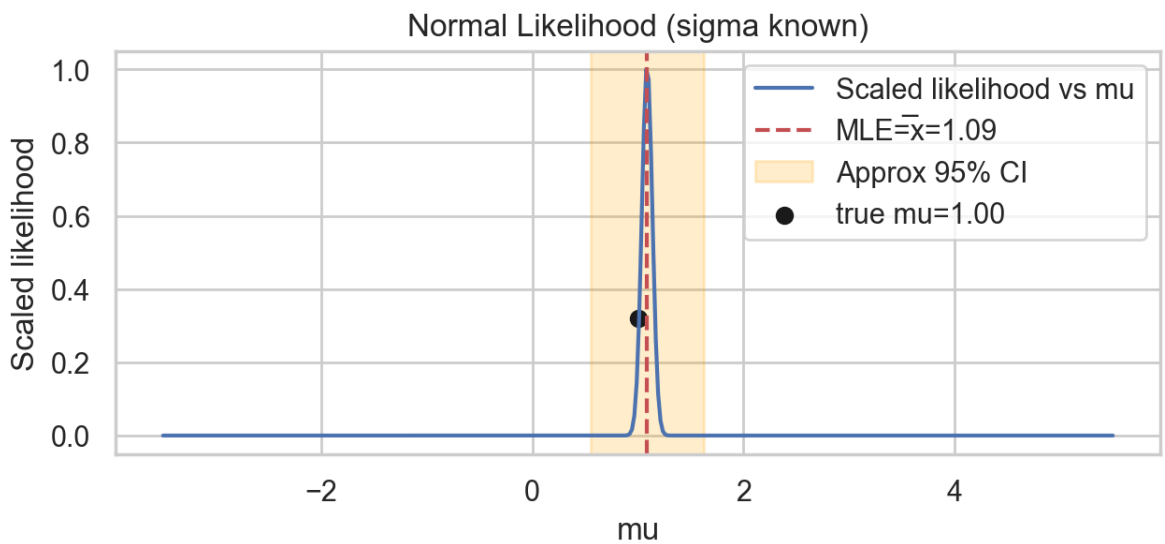

Figure 3: Scaled likelihood for μ (with σ known) showing a peak at $\hat{\mu} = \bar{x}$.

Solving this system yields:

$$\hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x} \tag{3}$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 \tag{4}$$

Note that $\hat{\sigma}^2_{MLE}$ differs from the unbiased sample variance by the factor $\frac{n-1}{n}$, illustrating that maximum likelihood estimators are not necessarily unbiased in finite samples.

4 Theoretical Properties of Maximum Likelihood Estimators

4.1 Invariance Property

One of the most useful properties of maximum likelihood estimators is their behavior under parameter transformations, which is particularly important in data science applications where parameters often need to be transformed for interpretation or computational convenience.

Theorem 4.1 (Invariance Property of MLE)**.** *Let $\hat{\theta}_{MLE}$ be the maximum likelihood estimator of θ, and let $g : \Theta \to \Gamma$ be a one-to-one function. Then $g(\hat{\theta}_{MLE})$ is the maximum likelihood estimator of $\gamma = g(\theta)$.*

Proof. Since g is one-to-one, there exists an inverse function g^{-1}. The likelihood function for γ is:

$$L^*(\gamma) = L(g^{-1}(\gamma))$$

Since $\hat{\theta}_{MLE}$ maximizes $L(\theta)$, and g is one-to-one, $g(\hat{\theta}_{MLE})$ maximizes $L^*(\gamma)$. □

This property is extremely useful in practice. For example, if $\hat{\sigma}^2_{MLE}$ is the MLE of the variance, then $\sqrt{\hat{\sigma}^2_{MLE}}$ is automatically the MLE of the standard deviation, without requiring separate optimization.

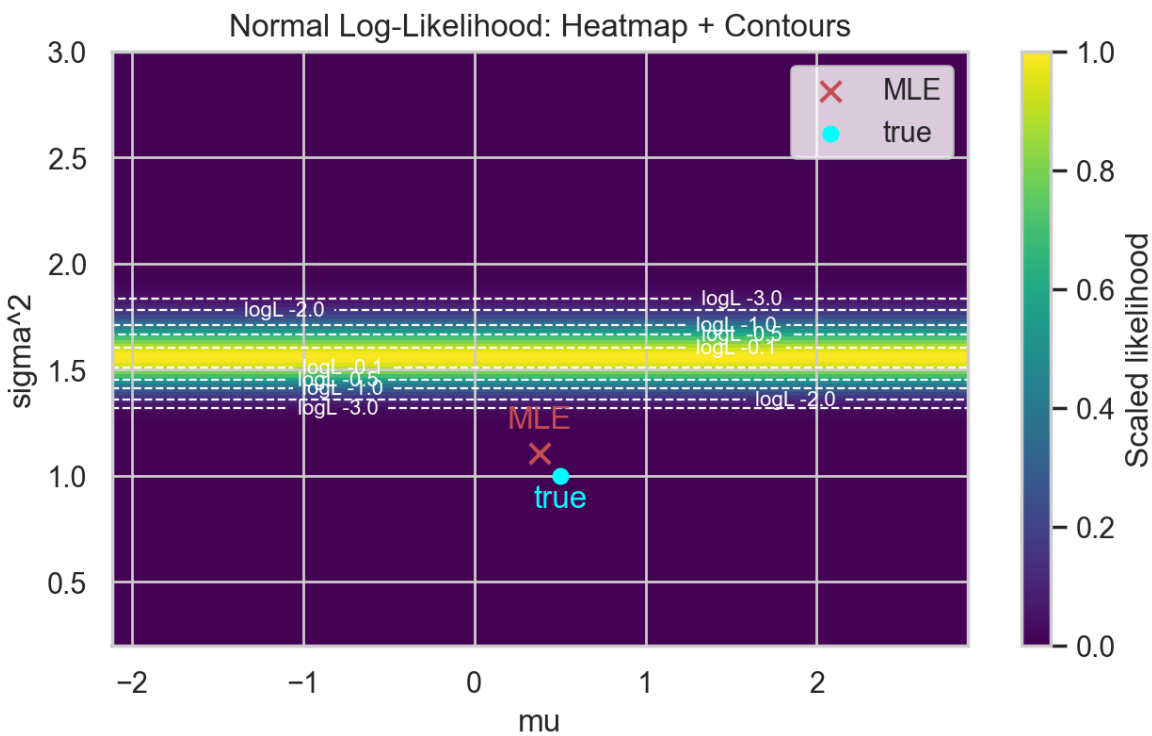

Figure 4: Contours of the normal log-likelihood over (μ, σ^2) with the MLE indicated.

4.2 Asymptotic Properties

The large-sample properties of maximum likelihood estimators provide the theoretical foundation for their widespread use in statistical inference and machine learning applications.

Theorem 4.2 (Asymptotic Properties of MLE)**.** *Under regularity conditions, as $n \to \infty$:*

1. ***Consistency**: $\hat{\theta}_{MLE} \xrightarrow{P} \theta_0$ (convergence in probability)*
2. ***Asymptotic Normality**: $\sqrt{n}(\hat{\theta}_{MLE} - \theta_0) \xrightarrow{D} N(0, I^{-1}(\theta_0))$*
3. ***Asymptotic Efficiency**: The MLE achieves the Cramér-Rao lower bound asymptotically*

where θ_0 is the true parameter value and $I(\theta)$ is the Fisher Information Matrix.

The regularity conditions required for these results include: the parameter space contains the true parameter in its interior, the likelihood function is twice differentiable, the Fisher information is finite and positive definite, and certain uniform convergence conditions hold. These conditions are satisfied by most common statistical models used in practice.

The consistency property ensures that MLEs converge to the true parameter values as sample sizes increase, providing theoretical justification for their use in large-sample applications. The asymptotic normality result enables the construction of confidence intervals and hypothesis tests based on the normal approximation, while the efficiency property guarantees optimal performance among all asymptotically unbiased estimators.

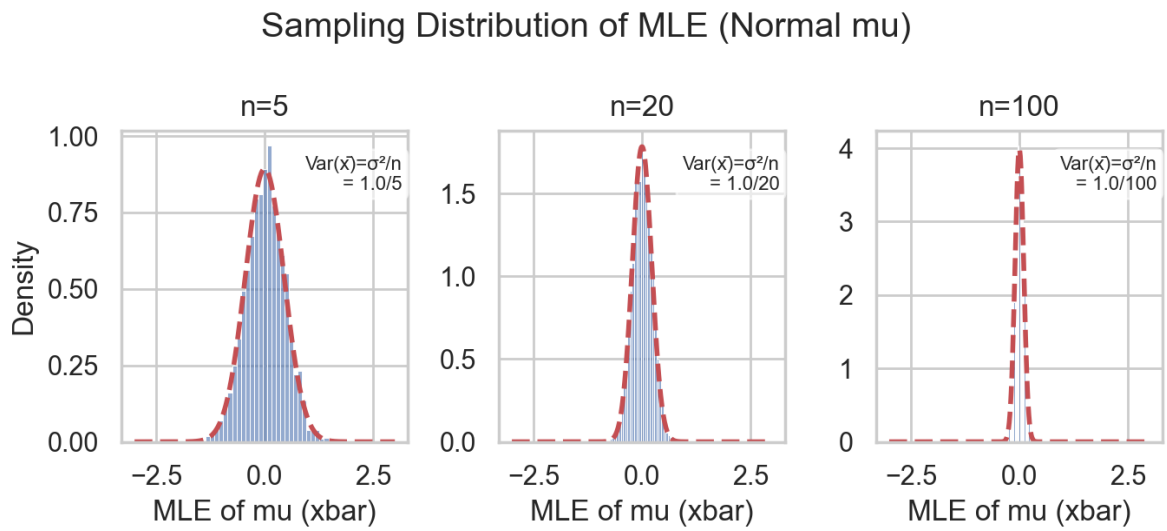

Figure 5: Sampling distribution of the MLE $\hat{\mu} = \bar{X}$ for increasing n with normal data: histogram overlaid with the corresponding normal approximation.

5 Fisher Information and Efficiency Theory

5.1 Fisher Information Matrix

The Fisher Information Matrix plays a central role in likelihood theory, quantifying the amount of information about parameters contained in the data and providing bounds on the precision of parameter estimation.

Definition 5.1 (Fisher Information Matrix)**.** *For a parameter vector $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_k)^T$, the Fisher Information Matrix is the $k \times k$ matrix with elements:*

$$I_{ij}(\boldsymbol{\theta}) = E\left[\frac{\partial \log f(X;\boldsymbol{\theta})}{\partial \theta_i}\frac{\partial \log f(X;\boldsymbol{\theta})}{\partial \theta_j}\right]$$

Under regularity conditions, the Fisher Information Matrix can also be expressed as:

$$I_{ij}(\boldsymbol{\theta}) = -E\left[\frac{\partial^2 \log f(X;\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}\right]$$

The Fisher Information Matrix is always positive semidefinite, and under identifiability conditions, it is positive definite. The inverse of the Fisher Information Matrix provides the asymptotic covariance matrix of the MLE, establishing fundamental limits on estimation precision.

5.2 Cramér-Rao Lower Bound

The Cramér-Rao lower bound establishes a fundamental limit on the variance of unbiased estimators, providing a benchmark for evaluating estimator performance.

Theorem 5.1 (Cramér-Rao Lower Bound)**.** *Let $\hat{\boldsymbol{\theta}}$ be any unbiased estimator of $\boldsymbol{\theta}$ based on a sample of size n. Under regularity conditions:*

$$Cov(\hat{\boldsymbol{\theta}}) \geq \frac{1}{n} I^{-1}(\boldsymbol{\theta})$$

where the inequality means that $Cov(\hat{\boldsymbol{\theta}}) - \frac{1}{n}I^{-1}(\boldsymbol{\theta})$ *is positive semidefinite.*

The Cramér-Rao bound provides a theoretical benchmark for comparing estimators. Estimators that achieve this bound are called efficient, and the MLE achieves this bound asymptotically under regularity conditions.

Definition 5.2 (Asymptotic Efficiency)**.** *An estimator sequence* $\{\hat{\boldsymbol{\theta}}_n\}$ *is asymptotically efficient if:*

$$\lim_{n\to\infty} n \cdot Cov(\hat{\boldsymbol{\theta}}_n) = I^{-1}(\boldsymbol{\theta})$$

The asymptotic efficiency of the MLE means that no other estimator can achieve smaller asymptotic variance, making the MLE optimal in large samples among all reasonable estimators.

6 Computational Methods and Optimization

6.1 Newton-Raphson Algorithm

When analytical solutions are not available, numerical optimization methods are essential for computing MLEs. The Newton-Raphson algorithm is particularly well-suited for likelihood maximization due to its use of second-derivative information.

Newton-Raphson Algorithm for MLE: Starting from an initial guess $\boldsymbol{\theta}^{(0)}$, iterate:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \left[\frac{\partial^2 \ell(\boldsymbol{\theta}^{(k)})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}\right]^{-1} \frac{\partial \ell(\boldsymbol{\theta}^{(k)})}{\partial\boldsymbol{\theta}}$$

until convergence.

The Newton-Raphson algorithm has quadratic convergence near the optimum, making it very efficient when good starting values are available. The algorithm naturally incorporates the curvature of the likelihood function through the Hessian matrix, leading to faster convergence than gradient-based methods.

6.2 Expectation-Maximization Algorithm

For models with latent variables or missing data, the Expectation-Maximization (EM) algorithm provides a powerful approach to maximum likelihood estimation that is guaranteed to increase the likelihood at each iteration.

Definition 6.1 (EM Algorithm)**.** *Let* $\mathbf{X}$ *denote the observed data and* $\mathbf{Z}$ *denote the latent variables. The EM algorithm alternates between:*

1. ***E-step****: Compute* $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)}) = E[\log L(\boldsymbol{\theta}|\mathbf{X},\mathbf{Z})|\mathbf{X},\boldsymbol{\theta}^{(k)}]$
2. ***M-step****: Set* $\boldsymbol{\theta}^{(k+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$

Theorem 6.1 (EM Convergence)**.** *The EM algorithm is guaranteed to increase the likelihood at each iteration:* $L(\boldsymbol{\theta}^{(k+1)}) \geq L(\boldsymbol{\theta}^{(k)})$*, with equality only at stationary points.*

The EM algorithm is particularly useful for mixture models, hidden Markov models, and other latent variable models that are common in machine learning applications.

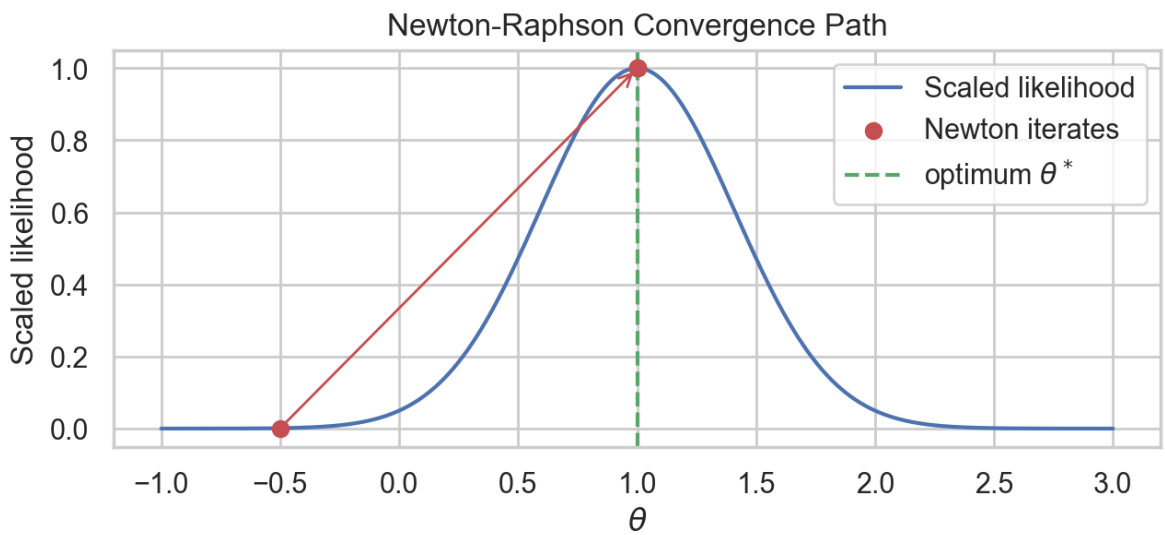

Figure 6: Newton-Raphson iterations on a concave log-likelihood proxy. Arrows show updates converging to the optimum.

7 Applications in Modern Data Science

7.1 Generalized Linear Models

Maximum likelihood estimation provides the foundation for generalized linear models (GLMs), which extend linear regression to non-normal response distributions while maintaining computational tractability.

For a GLM with response Y_i and predictors $\mathbf{x}_i$, the likelihood is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} f(y_i; \theta_i, \phi)$$

where $\theta_i = g^{-1}(\mathbf{x}_i^T \boldsymbol{\beta})$ and g is the link function.

Common examples include:

1. **Logistic Regression**: $g(\mu) = \log(\mu/(1-\mu))$ for binary responses
2. **Poisson Regression**: $g(\mu) = \log(\mu)$ for count responses
3. **Gamma Regression**: $g(\mu) = 1/\mu$ for positive continuous responses

7.2 Neural Networks and Deep Learning

Maximum likelihood estimation underlies the training of neural networks through the minimization of cross-entropy loss functions, which correspond to negative log-likelihood for appropriate distributional assumptions.

For classification with softmax outputs, the likelihood is:

$$L(\mathbf{W}) = \prod_{i=1}^{n} \prod_{k=1}^{K} p_{ik}^{y_{ik}}$$

where $p_{ik} = \frac{\exp(f_k(\mathbf{x}_i;\mathbf{W}))}{\sum_{j=1}^{K}\exp(f_j(\mathbf{x}_i;\mathbf{W}))}$ and $\mathbf{W}$ represents all network weights.

The negative log-likelihood becomes the cross-entropy loss function that is minimized during backpropagation training.

7.3 Mixture Models and Clustering

Gaussian mixture models provide a probabilistic approach to clustering that is naturally formulated in terms of maximum likelihood estimation.

For a mixture of K Gaussian components, the likelihood is:

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n}\sum_{k=1}^{K}\pi_k\phi(x_i;\mu_k,\Sigma_k)$$

where π_k are mixing proportions, μ_k are component means, and Σ_k are component covariance matrices.

The EM algorithm provides an elegant solution to this optimization problem, alternating between estimating cluster memberships (E-step) and updating component parameters (M-step).

8 Limitations and Robustness Considerations

8.1 Model Misspecification

Maximum likelihood estimation assumes that the specified model is correct, but real data often violate these assumptions. Model misspecification can lead to biased and inconsistent estimates, making robustness considerations important in practice.

When the true distribution differs from the assumed model, the MLE converges to the parameter value that minimizes the Kullback-Leibler divergence between the true and assumed distributions. This pseudo-true value may differ substantially from parameters of interest, leading to systematic bias in estimation and inference.

8.2 Computational Challenges

Modern data science applications often involve high-dimensional parameter spaces, large datasets, and complex model structures that create computational challenges for maximum likelihood estimation.

Local maxima can occur in non-convex likelihood surfaces, making the choice of starting values and optimization algorithms crucial for finding global optima. Regularization techniques such as ridge and LASSO penalties can help address overfitting and improve computational stability, though they modify the pure maximum likelihood criterion.

Numerical stability issues can arise when dealing with very small or large likelihood values, requiring careful implementation of log-sum-exp tricks and other numerical techniques to avoid underflow and overflow problems.

9 Summary

Maximum likelihood estimation provides a unified and theoretically elegant framework for parameter estimation that underlies much of modern statistical inference and machine learning.

The method combines intuitive appeal with rigorous mathematical foundations, achieving optimal asymptotic properties under mild regularity conditions.

The key insights from this material include the interpretation of likelihood as a measure of evidence about parameter values, the optimality properties of MLEs in large samples, and the central role of Fisher information in quantifying estimation precision. The computational methods developed for MLE, including Newton-Raphson and EM algorithms, have broad applications throughout data science.

The applications of MLE in generalized linear models, neural networks, and mixture models demonstrate its versatility and importance in modern data analysis. Understanding the limitations and robustness considerations is crucial for responsible application of these methods in practice.

The theoretical foundations covered here provide essential background for understanding more advanced topics in statistical learning, including Bayesian inference, regularization methods, and modern machine learning algorithms that build upon maximum likelihood principles.

10 Exercises

1. **Exponential Distribution MLE:** For independent observations $X_1, \ldots, X_n$ from an exponential distribution with parameter λ, derive the MLE $\hat{\lambda}_{MLE}$. Calculate the Fisher information $I(\lambda)$ and verify that the MLE achieves the Cramér-Rao lower bound.

2. **Gamma Distribution Complexity:** Consider the Gamma distribution with PDF $f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$. Derive the likelihood equations and explain why analytical solutions are not available. Describe a numerical approach using the Newton-Raphson algorithm.

3. **Invariance Property Application:** If $\hat{\sigma}^2_{MLE}$ is the MLE of the variance in a normal distribution, use the invariance property to find the MLE of the coefficient of variation $\gamma = \sigma/\mu$. Discuss the practical implications of this result.

4. **Multivariate Normal MLE:** For observations from a multivariate normal distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, derive the MLEs of both the mean vector and covariance matrix. Show that the MLE of the covariance matrix is biased and calculate the bias.

5. **Logistic Regression Likelihood:** For binary response data with logistic regression model $P(Y_i = 1) = \frac{1}{1+e^{-\mathbf{x}_i^T \boldsymbol{\beta}}}$, write out the log-likelihood function and derive the score equations. Explain why analytical solutions are not available and describe the iteratively reweighted least squares algorithm.

6. **EM Algorithm for Mixture Models:** For a two-component Gaussian mixture model, derive the E-step and M-step updates explicitly. Explain how the algorithm handles the latent variable representing component membership.

7. **Asymptotic Distribution Application:** Using the asymptotic normality of the MLE, construct a 95% confidence interval for the parameter λ of an exponential distribution. Compare this with exact confidence intervals based on the gamma distribution of the sufficient statistic.

8. **Likelihood Ratio Test Design:** For testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ in an exponential distribution, derive the likelihood ratio test statistic. Show that it follows a chi-square distribution under H_0 and calculate the power function for specific alternatives.